

Recall from a previous class:

- A set of vectors $\{\vec{v}_1, \dots, \vec{v}_p\}$ in $\mathbb{R}^n$ is said to be **linearly independent** if

$$x_1\vec{v}_1 + x_2\vec{v}_2 + \dots + x_p\vec{v}_p = \vec{0}$$

has only the trivial solution. The set is called **linearly dependent** if the equation has non-trivial solutions and the equation is called the **dependence relation**.

- Let $A = [\vec{a}_1 \dots \vec{a}_n]$, then $\{\vec{a}_1, \dots, \vec{a}_n\}$ are linearly independent if and only if $A\vec{x} = \vec{0}$ has only the trivial solution.
- **Theorem 7:** A set $S = \{\vec{v}_1, \dots, \vec{v}_p\}$ of two or more vectors is linearly dependent if and only if one of the vectors of S is a linearly combination of the others.

Moreover if $\vec{v}_1 \neq \vec{0}$, then some $\vec{v}_j$ for $j > 1$ is a linear combination of $\vec{v}_1, \dots, \vec{v}_{j-1}$.

- **Theorem 8:** Any set $\{\vec{v}_1, \dots, \vec{v}_p\}$ in $\mathbb{R}^n$ is linearly dependent if $p > n$.

Define a transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ to be a function from $\mathbb{R}^n$ to $\mathbb{R}^m$. Recall we call $\mathbb{R}^n$ the domain and $\mathbb{R}^m$ the codomain. The set of outputs, $\{T(\vec{x}) : \vec{x} \in \mathbb{R}^n\}$, is the image or the range of T .

One common example of a transformation T is given by $T(\vec{x}) = A\vec{x}$ where A is an m by n matrix

Definition 1. A transformation T is **linear** if

1. $T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$ for all $\vec{u}, \vec{v}$ in $\mathbb{R}^n$,
2. $T(c\vec{u}) = cT(\vec{u})$ for all $\vec{u}$ in $\mathbb{R}^n$ and c in $\mathbb{R}$.

1. Properties of Linear Transformations.

(a) Prove if T is linear, then $T(\vec{0}) = \vec{0}$.

(b) $T(c\vec{u} + d\vec{v}) = cT(\vec{u}) + dT(\vec{v})$.

2. Let $T(\vec{x}) = A\vec{x}$ where $A = \begin{bmatrix} 1 & 0 \\ -3 & 1 \\ 2 & -2 \end{bmatrix}$ and let $\vec{b} = \begin{bmatrix} -2 \\ 3 \\ -1 \end{bmatrix}$.

(a) Is $\vec{b}$ in the range of T ?

(b) What is the range of T ?

(c) Is T surjective?

(d) Is the transformation T injective?

3. Let $T(\vec{x}) = A\vec{x}$ where $A = \begin{bmatrix} 1 & 0 & -3 \\ -3 & 1 & 6 \\ 2 & -2 & -1 \end{bmatrix}$ and let $\vec{b} = \begin{bmatrix} -2 \\ 3 \\ -1 \end{bmatrix}$. Is $\vec{b}$ in the range of T ?

4. Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be linear and $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ be a linearly dependent set. Is $\{T(\vec{v}_1), T(\vec{v}_2), T(\vec{v}_3)\}$ linearly dependent?

5. Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be linear and $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ be a linearly **in**dependent set. Is $\{T(\vec{v}_1), T(\vec{v}_2), T(\vec{v}_3)\}$ linearly **in**dependent?

Linear Transformations – Answers / Solutions

1.

- (a) Suppose T is linear, then

$$T(\vec{0}) = T(0 \cdot \vec{u}) = 0 \cdot T(\vec{u}) = \vec{0}.$$

- (b) Suppose T is linear. By property (1) of linearity, we know

$$T(c\vec{u} + d\vec{v}) = T(c\vec{u}) + T(d\vec{v}).$$

By property (2), we know

$$T(c\vec{u}) + T(d\vec{v}) = cT(\vec{u}) + dT(\vec{v}).$$

Thus we have the result.

2.

- (a) In order to determine if $\vec{b}$ is in the range of T , we solve the matrix equation $A\vec{x} = \vec{b}$. Forming the augmented matrix and row-reducing, we get

$$\left(\begin{array}{cc|c} 1 & 0 & -2 \\ -3 & 1 & 3 \\ 2 & -2 & -1 \end{array} \right) \sim \left(\begin{array}{cc|c} 1 & 0 & -2 \\ 0 & 1 & -3 \\ 0 & 0 & -3 \end{array} \right).$$

Thus there is no solution to $A\vec{x} = \vec{b}$.

- (b) Replacing $\vec{b}$ for the previous part with arbitrary components will give the range of T . We get

$$\left(\begin{array}{cc|c} 1 & 0 & b_1 \\ -3 & 1 & b_2 \\ 2 & -2 & b_3 \end{array} \right) \sim \left(\begin{array}{cc|c} 1 & 0 & b_1 \\ 0 & 1 & b_2 + 3b_1 \\ 0 & 0 & b_3 + 2b_2 + 4b_1 \end{array} \right).$$

Thus $\vec{b}$ is in the range of T if and only if $b_3 + 2b_2 + 4b_1 = 0$. We can see that the range is a plane in $\mathbb{R}^3$.

- (c) No, T is not surjective. In particular, we showed in part (a) that there is no element in the domain that will map to $\vec{b} = \begin{bmatrix} -2 \\ 3 \\ -1 \end{bmatrix}$ under T . Equivalently we can say that $T(\vec{x}) = \vec{b}$ has no solutions.
- (d) Recall T is injective if for all $\vec{b}$, there is at most one $\vec{x}$ such that $T(\vec{x}) = \vec{b}$. We use the contrapositive form of injectivity. Suppose $T(\vec{x}) = T(\vec{y})$, then we get $A\vec{x} = A\vec{y}$. Rearranging the equation we get $A\vec{x} - A\vec{y} = \vec{0}$, and factoring, we get $A(\vec{x} - \vec{y}) = \vec{0}$. We want to show that $A\vec{z} = \vec{0}$ has only the trivial solution. Forming the augmented matrix and row-reducing, we find $A\vec{z} = \vec{0}$ has only the trivial solution. Therefore $A(\vec{x} - \vec{y}) = \vec{0}$ if and only if $\vec{x} - \vec{y} = \vec{0}$, and hence $\vec{x} = \vec{y}$. Thus T is injective.

3. Yes, $\vec{b}$ is in the range of T . In order to determine if $\vec{b}$ is in the range of T , we solve the matrix equation $A\vec{x} = \vec{b}$. Forming the augmented matrix and row-reducing, we get

$$\left(\begin{array}{ccc|c} 1 & 0 & -3 & -2 \\ -3 & 1 & 6 & 3 \\ 2 & -2 & -1 & -1 \end{array} \right) \sim \left(\begin{array}{ccc|c} 1 & 0 & 0 & 7 \\ 0 & 1 & 0 & 6 \\ 0 & 0 & 1 & 3 \end{array} \right).$$

Thus there is a unique solution to $A\vec{x} = \vec{b}$.

4. Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be linear and $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ be a linearly dependent set. Since $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ is linearly dependent, we know there exists c_1, c_2, c_3 , not all zero, such that $c_1\vec{v}_1 + c_2\vec{v}_2 + c_3\vec{v}_3 = \vec{0}$. Applying T to both sides of the equation we get

$$T(c_1\vec{v}_1 + c_2\vec{v}_2 + c_3\vec{v}_3) = T(\vec{0}).$$

Using linearity of T , we get

$$c_1T(\vec{v}_1) + c_2T(\vec{v}_2) + c_3T(\vec{v}_3) = \vec{0}.$$

Since not all c_1, c_2, c_3 are zero, we have a non-trivial solution to the vector equation

$$x_1T(\vec{v}_1) + x_2T(\vec{v}_2) + x_3T(\vec{v}_3) = \vec{0},$$

and hence $\{T(\vec{v}_1), T(\vec{v}_2), T(\vec{v}_3)\}$ is linearly dependent.

5. No. For example the transformation $T(\vec{x}) = A\vec{x}$ where A is the m by n matrix with all entries equal to zero is a linear transformation. However $\{T(\vec{v}_1), T(\vec{v}_2), T(\vec{v}_3)\} = \{\vec{0}, \vec{0}, \vec{0}\}$ is a linearly dependent set for any vectors $\vec{v}_i$.